Real-Time Sign Language to Speech Translator

Bridging Communication Gaps with AI

Team Name : Hackstorm

Team Member : Mayank, Aman , Aniket & Bittu

Event : HackGenesis

Made with GAMMA

Our Idea

→Build a system that understands **Indian Sign Language gestures.**

→Use a **webcam + AI (Computer Vision)** to detect hand movements.

→Instantly convert gestures into:

📝 Text ,🔊 Speech

Problem

→Deaf and hard-of-hearing people struggle to communicate in everyday situations.

→Most apps and systems are designed for voice or text, not sign language.

→Indian Sign Language (ISL) still lacks real-time translation tools.

Made with GAMMA

Solution

→We developed a **real-time ISL translation system.**

→Uses **webcam input** to capture hand gestures.

→Applies computer vision + AI model to recognize signs.

→Convert grstures into:

- 🔊Speech output(TTS)
- 📝 Text output

Made with GAMMA

Features

→Real-Time Gesture Recognition

→Accurate Hand Tracking (MediaPipe)

→Instant Text Conversion

→Speech Output (Text-to-Speech)

→Works with Standard Webcam

→Fast & User-Friendly Interface

→Speaking Version (natural flow)

Made with GAMMA

Tech Stack

→**Programming Language:** Python.

→**Computer Vision:** OpenCV.

→**Hand Tracking:** MediaPipe.

→**Machine Learning:** TensorFlow / Custom Model.

Made with GAMMA

Thank you...

Made with GAMMA